PROPER SMOOTH PUSHFORWARDS OF UNIFORMLY TORSION LOCALLY CONSTANT SHEAVES ARE FINITE LOCALLY CONSTANT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Constructibility of proper pushforwards (Theorem theorem-constructible-shriek, canonical tag 0GL0). cited at proof lines 5064-5065. In the final Noetherian n-torsion case it provides constructibility of the sheaves R^if_*F, the input to the specialization and characterization lemmas.
- Locally constant sheaves as a colimit (Etale Cohomology, Lemma etale-cohomology-lemma-category-loc-cst-is-colimit, canonical tag 0GL2). cited at proof lines 5051-5055. Lets the locally constant sheaf F descend to a finite type locally constant F_i on some X_i in the Noetherian descent.
- Locally constant characterization (Etale Cohomology, Lemma etale-cohomology-lemma-characterize-locally-constant-module, canonical tag 0GKC). cited at proof lines 5069-5070. The terminal criterion: constructible sheaves whose specialization maps are isomorphisms are locally constant.

Source reference. Original lemma 0GL2: sources/stacks-chow/tags.txt, source lines 20707–20707. Cited at proof lines 5051-5055: F descends to a finite type locally constant sheaf F_i on some X_i.

Source reference. Original lemma 0GKC: sources/stacks-chow/tags.txt, source lines 20682–20682. Cited at proof lines 5069-5070: constructible sheaves whose specialization maps are isomorphisms are locally constant, finishing the proof.

Source reference. Original lemma 0GJW: sources/stacks-chow/tags.txt, source lines 20666–20666. Cited at proof lines 5066-5068: the specialization maps of R^if_*F are isomorphisms.

Source reference. Original lemma 0EY2: sources/stacks-chow/tags.txt, source lines 18677–18677. Cited at proof lines 5046-5050: one of the three Limits lemmas used to improve f_i after increasing i.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cited at proof lines 5042-5045: descend f to a finitely presented morphism f_i : X_i -> S_i over a finite type Z[1/n]-subalgebra.

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Source reference. Original lemma 0C0C: sources/stacks-chow/tags.txt, source lines 15082–15082. Cited at proof lines 5046-5050: after increasing i, f_i is smooth.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cited at proof lines 5046-5050: after increasing i, f_i is proper.

Source reference. Original theorem 0GL0: sources/stacks-chow/tags.txt, source lines 20705–20705. Cited at proof lines 5064-5065: the sheaves R^if_*F are constructible in the Noetherian n-torsion case.

Proposition 0.2. *Let $f : X \to S$ be a smooth proper morphism of schemes. Let Λ be a Noetherian ring. Let $\mathcal{F}$ be a finite type, locally constant sheaf of Λ-modules on $X_{étale}$ such that for every geometric point $\overline{x}$ of X the stalk $\mathcal{F}_{\overline{x}}$ is annihilated by an integer $n > 0$ prime to the residue characteristic of $\overline{x}$. Then $R^i f_* \mathcal{F}$ is a finite type, locally constant sheaf of Λ-modules on $S_{étale}$ for all $i \in \mathbf{Z}$.*

Proof. The question is local on S and hence we may assume S is affine. For a point x of X denote $n_x \geq 1$ the smallest integer annihilating $\mathcal{F}_{\overline{x}}$ for some (equivalently any) geometric point $\overline{x}$ of X lying over x. Since X is quasi-compact (being proper over affine) there exists a finite étale covering $\{U_j \to X\}_{j=1,\ldots,m}$ such that $\mathcal{F}|_{U_j}$ is constant. Since $U_j \to X$ is open, we conclude that the function $x \mapsto n_x$ is locally constant and takes finitely many values. Accordingly we obtain a finite decomposition $X = X_1 \amalg \ldots \amalg X_N$ into open and closed subschemes such that $n_x = n$ if and only if $x \in X_n$. Then it suffices to prove the lemma for the induced morphisms $X_n \to S$ and the restriction of $\mathcal{F}$ to X_n. Thus we may and do assume there exists an integer $n > 0$ such that $\mathcal{F}$ is annihilated by n and such that n is prime to the residue characteristics of all residue fields of X.

Since f is smooth and proper the image $f(X) \subset S$ is open and closed. Hence we may replace S by $f(X)$ and assume $f(X) = S$. In particular, we see that we may assume n is invertible in the ring defining the affine scheme S.

In this paragraph we reduce to the case where S is Noetherian. Write $S = \mathrm{Spec}(A)$ for some $\mathbf{Z}[1/n]$-algebra A. Write $A = \bigcup A_i$ as the union of its finite type $\mathbf{Z}[1/n]$-subalgebras. We can find an i and a morphism $f_i : X_i \to S_i = \mathrm{Spec}(A_i)$ of finite type whose base change to S is $f : X \to S$, see Limits, Lemma 01ZM. After increasing i we may assume $f_i : X_i \to S_i$ is smooth and proper, see Limits, Lemmas 081F 0C0C, and 0EY2. By Étale Cohomology, Lemma 0GL2 we see that there exists an i and a finite type, locally constant sheaf of Λ-modules $\mathcal{F}_i$ whose pullback to X is isomorphic to $\mathcal{F}$. As $\mathcal{F}$ is annihilated by n, we may replace $\mathcal{F}_i$ by $\mathrm{Ker}(n : \mathcal{F}_i \to \mathcal{F}_i)$ and assume the same thing is true for $\mathcal{F}_i$. This reduces us to the case discussed in the next paragraph.

Assume we have an integer $n \geq 1$, the base scheme S is Noetherian and lives over $\mathbf{Z}[1/n]$, and $\mathcal{F}$ is n-torsion. By Theorem 0GL0 the sheaves $R^i f_* \mathcal{F}$ are constructible sheaves of Λ-modules. By Étale Cohomology, Lemma 0GJW the specialization maps of $R^i f_* \mathcal{F}$ are always isomorphisms. We conclude by Étale Cohomology, Lemma 0GKC. □